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What Makes a Robot Feel Affective? Verbal Cue Types and the Role of Agreeableness in Older Adults' Perception of Robot Personality

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Abstract

This study explores the potential of social robots with distinct personality profiles for older adult companionship, focusing on the legibility of affective verbal cues during initial encounters, an area presently underexplored. In a within-subjects experiment, 30 Mandarin-speaking older adults ($M_{\text{age}} = 72.4$) interacted with two LLM-driven robots exemplifying high-affective (ENFP) and low-affective (ENTP) personalities. Linguistic distinctions were validated through LIWC analysis of conversational turns. Results showed that 80% of participants accurately identified the high-affective robot's personality. Exploratory factor analysis of perception ratings identified two affective cue subtypes: Feeling Type I (other-directed emotional resonance), which effectively conveyed affectiveness, and Feeling Type II (self-referential emotional volatility), which impaired it, aligning with thin-slice judgment theory's differentiation between expressive external cues and internal states. Agreeableness emerged as the only Big Five predictor of preference for the high-affective robot (Cohen's $d = 0.86\text{--}1.16$). These findings suggest that other-directed emotional expression is a legible and viable target for affective human–robot interaction in single-session settings with older adults, whereas self-referential cues may carry risks. Pending further validation with larger samples, these insights can inform design strategies for LLM-based companion robots within the CASA paradigm, emphasizing affective expressiveness that enhances clarity during initial exchanges.

Keywords: Robot affectiveness; older adults; verbal cues; agreeableness; personality robots

1. Introduction

Consider the opening minutes of a social robot's first conversation with an older adult. Within seconds, the older adult immediately begins forming impressions: is this robot warm or cold, emotionally present or mechanically detached? For robot designers working with LLM-driven conversational systems, a sharp practical question emerges: which verbal expressions of affectiveness are immediately legible in this first encounter, and which, despite being programmed as emotionally expressive, may inadvertently signal something the older adult finds unsettling rather than comforting? This question draws on thin-slice judgment research, social robotics, and gerontechnology design, yet has received remarkably little empirical attention.

The broader field of robot personality design has made substantial progress in demonstrating that social robots can meaningfully vary along personality dimensions, and that these variations influence human perception and interaction quality (Mou et al., 2020). However, the majority of this work has focused on dimensions such as extraversion and dominance, leaving affective characteristics comparatively underexplored (Picard, 1997). In this study, affective characteristics are defined as the degree to which a robot expresses emotional resonance, care for others' feelings, and intuitive responsiveness. Yet existing research has overwhelmingly recruited younger participants, leaving a significant gap in understanding how older adults, who possess distinct interaction scripts and heightened needs for emotional companionship (Carstensen et al., 1999), perceive and respond to robots with different degrees of affectiveness.

This study addresses existing research gaps through a within-subjects experimental design in which older adults engaged in conversations with two LLM-driven robots that embodied contrasting personality profiles. It empirically identifies two qualitatively distinct affective verbal cue subtypes that demonstrated opposing correlations with perceptions of affectiveness in single-session human-robot interaction, a distinction grounded in the differential accessibility of outward expressive cues versus internal states, as outlined by Ambady and Rosenthal (1992) and Carney et al. (2007). Furthermore, the study tests the *a priori* hypothesis that agreeableness, defined as individual differences in empathy, cooperation, and prosocial orientation (Graziano & Eisenberg, 1997), predicts older adults' preference for high-affective robots, thereby examining whether personality-driven interpersonal preference scripts are transferable from human-human to human-robot contexts (Gambino et al., 2020; Nass et al., 1994). As a manipulation check, this study verifies that older adults, characterized by distinct interaction scripts and heightened emotional needs (Carstensen et al., 1999), can distinguish between high- and low-affective robot personalities within a single conversational session. Framing these objectives within the theoretical contexts of thin-slice judgment theory (Ambady & Rosenthal, 1992; Carney et al., 2007), the computer as social actor (CASA) paradigm (Nass et al., 1994; Reeves & Nass, 1996), and older adults' HRI interaction scripts (Chang et al., 2018; Martínez-Miranda et al., 2018), the study aims to provide empirically grounded design recommendations for emotionally adaptive companion robots tailored to older adult populations.

2. Background

2.1. Thin-Slice judgments and affective verbal cues in single-session HRI

Thin-slice judgment research demonstrates that humans form accurate personality impressions from remarkably brief behavioral samples, often within seconds of an initial encounter (Ambady & Rosenthal, 1992). Critically for social robotics, not all personality dimensions are equally accessible in thin slices. Carney et al. (2007) established a differential accessibility

hierarchy: extraversion and conscientiousness can be reliably perceived within seconds, while positive affect, neuroticism, openness, and agreeableness require longer interaction periods or more specific behavioral cues. This differential accessibility explains why conversational tone and immediate feedback style serve as diagnostically powerful cues for some traits, whereas internal states such as self-doubt or emotional vulnerability require extended observation to be reliably perceived.

Research on first impressions in HRI has extended thin-slice findings to robot evaluation contexts. Carpinella et al. (2017) demonstrated in large-scale studies that warmth and competence, the two primary dimensions of the Stereotype Content Model, are as central to robot evaluation as to person evaluation, with impressions forming rapidly from appearance and early conversational behavior alone. Paetzel-Prüsmann et al. (2021) further showed that robots rated highly on agreeableness-related characteristics can weaken uncanny feelings arising from initial impressions, a result that confirms affective verbal cues carry particularly high diagnostic weight in short-duration HRI. The process by which people ascribe humanness to conversational agents has been theorized as a dynamic attribution process driven by perceived responsiveness, consistency, and the congruence between conversational style and social expectations (Rapp et al., 2024). That account aligns well with the thin-slice judgment framework and further motivates examining verbal cue types as the unit of analysis. No published study has specifically examined which categories of affective verbal behavior are most legible within a single conversational session with older adult users. This gap is particularly consequential given that older adults typically have limited prior experience with robots, and their impression formation processes may differ substantially from those of younger cohorts (Rossi et al., 2024).

2.2. Robot personality design and the MBTI framework

Research on robot personality design has traditionally relied on the Big Five personality model as both a design and measurement framework, given its well-established psychometric properties and cross-cultural validity (Schmitt et al., 2007). A century-long bibliometric analysis of robot personality descriptions confirms that agreeableness has consistently ranked among the most prominent Big Five dimensions in cultural representations of robots, second only to openness throughout most of the twentieth century and increasingly central in the social robotics era (Xu & Chao, 2024). However, the Big Five model presents a specific challenge for operationalizing robot behavior: its dimensions, while theoretically coherent, do not map directly onto discrete verbal behavior categories that can be programmed into a conversational robot. Agreeableness, for instance, is defined at the trait level as a disposition toward cooperation and empathy, but translating this into concrete robot utterances requires bridging the gap between trait description and behavioral expression in ways that the Big Five framework itself does not specify.

The Myers-Briggs Type Indicator (MBTI), and specifically its Feeling–Thinking (F–T) dimension, addresses this operationalization challenge by providing a more behaviorally specific contrast. The Feeling type is characterized by explicit verbal behaviors: prioritizing others’ emotional states in decision-making, expressing empathic resonance, and using intuitive rather than analytic reasoning in conversation. The Thinking type, conversely, is characterized by a prioritization of facts, logical consistency, and efficiency over emotional considerations (Myers et al., 1998). This behavioral specificity makes the F–T dimension particularly well-suited to prompt engineering for LLM-driven robots: the contrast between ENFP (high-affective) and ENTP (low-affective) personalities can be instantiated through specific verbal behavior guidelines that are interpretable by the LLM and consistent across conversational topics.

Furthermore, extensive research has identified consistent linguistic features derived from naturalistic data sources such as social media posts, which pertain to specific dimensions of the Myers-Briggs Type Indicator (MBTI) (Gjurković & Šnajder, 2018; Plank & Hovy, 2015). Specifically, Plank and Hovy (2015) demonstrated that the thinking–feeling dimension exhibits distinctive and stable linguistic markers. Their stability-selection analysis indicated that individuals with thinking profiles tend to use markers indicative of propositional certainty and pragmatic or task-oriented contexts, whereas those with feeling profiles are more likely to employ interpersonal and explicit affective expressions. Prior studies further support that the thinking-feeling axis produces divergent, measurable linguistic features in natural language, thereby suggesting that verbal stimuli designed along this dimension can be computationally validated.

Recent research on LLM personality alignment provides further support for this approach. Cui et al. (2023) demonstrated stable MBTI-type alignment across multiple LLM architectures through fine-tuning and direct preference optimization, suggesting that MBTI types can be reliably instantiated in LLM-driven conversational systems. Besta et al. (2025) proposed a lightweight MBTI-based framework for controlling LLM agent behavior along cognitive and affective axes without requiring full fine-tuning. Together, these developments validate the use of MBTI as a robot design framework in LLM-driven systems, independent of the psychometric concerns raised regarding its use as a human personality assessment tool.

In the present study, MBTI serves exclusively as a robot design framework, specifying the verbal behavior categories that differentiate the two robot personalities. Participant personality is assessed using the Big Five Inventory (BFI-15), which retains the psychometric properties necessary for examining individual-difference predictors of robot preference. This dual-framework approach, employing MBTI for operationalizing robot design and the Big Five for participant characterization, reflects the distinct functional requirements of each assessment context.

2.3. Agreeableness and social interaction preferences

Agreeableness represents individual differences in cooperation, empathy, and prosocial behavior, and has emerged as particularly relevant to emotional interactions and preferences (Graziano & Eisenberg, 1997). High-agreeableness individuals demonstrate enhanced empathic responses (Graziano et al., 2007), stronger prosocial tendencies (Habashi et al., 2016), and greater sensitivity to others' emotional states. These characteristics should, in principle, make them more attuned to outward emotional cues in social interactions.

Research consistently identifies agreeableness as the strongest predictor of empathy quotient across cultures and measurement instruments (Wakabayashi & Kawashima, 2015). The underlying mechanism involves superior perspective-taking abilities: low-agreeableness individuals report less empathy not because they lack empathic motivation, but because they show reduced skill in shifting attentional focus to others' emotional states (Graziano et al., 2007). In the thin-slice judgment context, this suggests that highly agreeable individuals should show enhanced sensitivity specifically to other-directed affective verbal cues that directly signal emotional resonance, compared with less agreeable individuals.

The distinction between other-directed and self-referential emotional expression is deeply rooted in the empirical literature on empathy. Research in this domain distinguishes between other-oriented empathic concern and self-oriented personal distress; the former pertains to attending to and responding to others' emotional states, whereas the latter involves self-focused emotional arousal that may impede prosocial behavior (Eisenberg & Miller, 1987). This bipartite construct is firmly embedded in standard psychometric tools, which typically measure

empathic concern and personal distress as separate factors (Chrysikou & Thompson, 2016). The attentional mechanisms associated with agreeableness directly correspond to this dichotomy: individuals exhibiting high levels of agreeableness are characterized by their enhanced ability to shift attentional focus toward others' emotional cues (Graziano et al., 2007), thereby rendering such affective signals especially salient. At the behavioral level, this distinction manifests in language use: profiles characterized by affective states are marked by interpersonal and explicitly emotional expressions (Plank & Hovy, 2015), while self- versus other-focus can be identified through specific linguistic patterns (Fast & Funder, 2008). Collectively, the above-mentioned literature suggests that other-directed and self-referential emotional expressions represent psychologically and linguistically distinct categories of cues, which this study empirically investigates in the context of robotic verbal behavior.

In healthcare contexts, agreeableness correlates with better patient outcomes and provider performance (Esterwood & Robert, 2021). Kabacińska et al. (2025) confirmed in a systematic review of 56 HRI studies that agreeable participants showed greater comfort and engagement with social robots. Paetzel-Prüsmann et al. (2021) additionally demonstrated that agreeableness-related robot characteristics can modulate uncanny valley responses, suggesting that affective robot design specifically activates the empathy-related personality systems associated with agreeableness rather than broader social trait patterns. These characteristics suggest that highly agreeable older adults may be particularly responsive to robots exhibiting other-directed affective verbal behaviors, as such verbal cues align with their interpersonal orientation and social harmony preferences. This pattern resonates with evidence that individual differences in trust and social sensitivity shape differentiated responses across robot interaction conditions (Yueh et al., 2026b).

2.4. Human–robot interaction scripts and older adults

The Computers as Social Actors (CASA) paradigm provides the foundational theoretical framework for understanding how personality traits influence human-robot interactions, demonstrating that humans instinctively apply social rules to computer systems despite awareness of their non-human nature (Nass et al., 1994; Reeves & Nass, 1996). This framework, extended by Gambino et al. (2020) to account for unique human-media interaction scripts, helps explain why individual differences in agreeableness may lead to distinct preferences for robot personality types. Recent replication work has qualified the scope of CASA: Heyselaar (2023) found that CASA effects no longer hold for desktop computers under direct replication conditions, suggesting that social response patterns may be specific to technologies perceived as genuinely social agents. This qualification strengthens rather than undermines the application of CASA to embodied social robots, particularly for older adult users, who typically lack the technology-specific interaction scripts that attenuate social responding among experienced users (Gambino et al., 2020). For this population, the robot remains a genuinely novel, socially appearing entity, and CASA effects are therefore expected to be robust and potentially amplified.

Research reveals that older adults develop specific interaction scripts when engaging with conversational robots, distinct from those used in human interactions and other technology use (Chang et al., 2018; Martínez-Miranda et al., 2018). Regarding conversational expectations, older adults anticipate bidirectional dialogue in which both parties meaningfully respond to each other, and most find single-directional exchanges unsatisfying or jarring (Ho et al., 2018; Li & Larivière, 2023). Regarding emotional expression, preferences vary considerably: some older adults favor highly expressive emotional displays, while others prefer subtle nonverbal cues that signal attention and engagement (de Visser et al., 2016). However, a near-universal expectation across older adult participants is that robots should display emotional stability and

reliability. This expectation is rooted in the practical context of companion robot use, where consistency and predictability are valued over emotional expressiveness per se.

Most significantly for the present study, older adults frequently anthropomorphize robots, assigning them familial roles and developing emotional bonds over sustained interactions (Epley et al., 2007; Nowak & Biocca, 2003). This personification process directly affects the depth of emotional engagement in HRI, with sustained two-way communication enhancing emotional bonding over time (Biocca et al., 2003; Gaudiello et al., 2016). However, expectations of robot stability and competence remain central to this personification process: attributes that violate these expectations, such as self-doubt, volatility, or rumination, are likely to disrupt the anthropomorphization process and trigger negative evaluations, regardless of whether the robot also displays affective warmth. This account predicts that inward-directed robot behaviors such as self-doubt, volatility, or rumination would be particularly disruptive in this population, as they conflict with both the stability expectation and the process of anthropomorphization. Complementary design-based research confirms that proactive and memory-aware conversational strategies improve older adult engagement with library service robots (Yueh et al., 2026a), while error-handling approaches in chatbot dialogue have been shown to significantly affect older users' interaction quality and task completion in technology-learning contexts (Lin et al., 2021).

Despite this theoretical foundation, the specific context of Taiwan is directly relevant here: it shares both the demographic profile and cultural background of the present study. Recent IJHCI research confirms that hedonic and utilitarian attitudes, alongside conscientiousness, shape caregivers' behavioral intention to adopt socially assistive AI robots for their elderly family members, underscoring the practical relevance of personality-related factors in this population's technology decisions (Lin et al., 2024). Nevertheless, the specific verbal behaviors that satisfy or violate older adults' interaction scripts within a single conversational session, and the role of individual personality traits in moderating these responses, have not been empirically characterized in prior research. This gap is the primary motivation for the present investigation.

3. Method

3.1. Participants

Thirty native Mandarin-speaking older adults were recruited from community centers in southern and northern Taiwan. Participants underwent screening for hearing difficulties and Mandarin comprehension. Final data from 30 participants (23 females; mean age = 72.37 years; mean years of education = 13.27; age range 63–84 years) were included in analyses. Participants demonstrated adequate baseline technology acceptance (Technology Acceptance Model mean score = 64.10), indicating that technology anxiety was unlikely to substantially confound the results. Table 1 presents the complete participant demographics and Big Five Inventory scores. All participants provided written informed consent and received financial compensation. All experimental procedures were approved by the Institutional Review Board of National Taiwan University.

Table 1.
Participant demographics and BFI-15 scores.

Variables	Descriptions
Sex	23 females, 7 males

Age, M (SD)	72.37 (5.2); range 63–84
Education (years), M (SD)	13.27 (3.46); range 6–18
TAM total score, M (SD)	64.10 (11.41)
BFI-15 — Openness, M (SD)	10.97 (2.55)
BFI-15 — Conscientiousness, M (SD)	12.83 (1.88)
BFI-15 — Extraversion, M (SD)	12.23 (2.22)
BFI-15 — Agreeableness, M (SD)	12.33 (1.86)
BFI-15 — Neuroticism, M (SD)	8.47 (3.05)

3.2. Research hypotheses and experimental design

This study tested two primary hypotheses. Hypothesis 1 (H1: Manipulation Check) posits that older adults can distinguish between high- and low-affective robot personalities through conversational human-robot interaction when interaction scripts designed to promote bidirectional dialogue, emotional expression, and anthropomorphism are employed. Building upon previous research demonstrating that the current robotic system effectively implements these scripts (Lo et al., 2025), it was hypothesized that participants would accurately identify robot personalities at levels exceeding chance. Hypothesis 2 (H2: Primary Research Question) predicts that individual differences in agreeableness modulate older adults' preferences regarding the affective expressiveness of robots, with higher levels of agreeableness predicting a preference for highly affective robots. This hypothesis is consistent with the CASA paradigm (Gambino et al., 2020; Nass et al., 1994), while H1 draws on thin-slice judgment research demonstrating that trait impressions form reliably from brief behavioral samples (Ambady & Rosenthal, 1992; Carney et al., 2007).

The experimental design employed a within-subjects approach in which each participant interacted with both robot types in counterbalanced order (Figure 1). Prior to robot interactions, participants completed demographic questionnaires, the Technology Acceptance Model (TAM) scale, and the Big Five Inventory (BFI-15). Each participant then engaged in two separate 15-minute conversations: one with a robot programmed to exhibit ENFP (high-affective) personality traits, and one with a robot programmed to exhibit ENTP (low-affective) traits. The interaction sequence began with casual discussions about daily life, progressed through randomly assigned topics including career choices, family relationships, and personal experiences, and concluded naturally. Following each robot interaction, participants completed the MBTI-based affectiveness assessment scale. After both interactions, participants provided qualitative personality descriptions of each robot and indicated their preference between the two.

Sample size adequacy was evaluated through a priori power analysis using G*Power 3.1 (Faul et al., 2007) for a within-subjects repeated-measures design with two conditions. Based on the effect sizes subsequently observed (Cohen's $d = 0.856$ for high- vs. low-affective preference comparisons), a correlation of $\rho = 0.50$ between repeated measures, $\alpha = 0.05$, and desired power of 0.80, the minimum required sample size was estimated at $N = 23$. The present sample of $N = 30$ thus provided an estimated statistical power of approximately 0.90 for primary within-subjects comparisons. For the multinomial logistic regression analyses, where the sample-to-predictor ratio is more constrained, robustness was verified via bootstrap resampling across 2,000 iterations, as detailed in Section 4.

3.3. Measures

3.3.1. *Technology acceptance model (TAM)*

The Mandarin Chinese version of the TAM (Lai et al., 2015) is a seven-point Likert-type scale comprising 11 items across three domains: Perceived Usefulness, Perceived Ease of Use, and Behavioral Intention. The scale demonstrates strong internal consistency (Cronbach's $\alpha = 0.918$; Adams et al., 1992). In the present study, TAM scores are used as a participant background characteristic to confirm adequate technology acceptance across the sample, rather than as a predictor variable in the primary analyses.

3.3.2. *Big five inventory—short form (BFI-15)*

The Mandarin Chinese BFI-15 (Li & Chung, 2020) is a five-point Likert-type scale comprising 15 items across five domains: Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism. Internal consistency for the BFI-15 subscales ranges from $\alpha = 0.44$ to 0.76 (Hahn et al., 2012). Although lower than full-form Big Five inventories, the BFI-15 was selected given the physical endurance considerations of elderly participants and its established criterion-related validity with the BFI-44. Higher scores indicate stronger endorsement of the respective personality trait.

3.3.3. *MBTI-based affectiveness scale*

A 19-item scale derived from the Myers-Briggs Type Indicator Feeling-Thinking dimension was developed to assess participants' perceptions of each robot's affectiveness following each interaction session. Items were presented on a seven-point Likert scale; higher scores indicate a predominantly affectiveness-oriented perception of the robot's personality, while lower scores reflect a more thinking-oriented perception. The factorial structure of the scale was examined through Principal Component Analysis (PCA) with oblique rotation, conducted on 60 observations (30 participants $\times$ 2 robot conditions). The Kaiser-Meyer-Olkin measure of sampling adequacy was .69. While this value approaches the conventional acceptable threshold of 0.70 (Kaiser, 1974), given the present sample size, it is considered marginally adequate for exploratory factor analysis. This limitation is acknowledged, and the three-factor solution should be interpreted with appropriate caution pending replication with larger samples. The factor structure, item retention procedure, and final subscale composition are reported in Section 4.1.

3.4. Social robot system

The HRI system comprises a Python-based cognition server and a Java-based client communicating via APIs. The robot cognition server integrates LLM and Retrieval-Augmented Generation (RAG) to generate personalized dialogues, emotions, and actions. The system is implemented on the Kebbi robot® from Nuwa Robotics, which features 12 degrees of freedom, five sensing areas, an Android tablet (Android 9, KebbiOS 11.2), cameras, microphones, and speakers. The Kebbi HRI Interface manages automatic speech recognition (ASR), text-to-speech (TTS), message transmission, and synchronized execution of physical gestures and facial expressions.

Kebbi was deployed with personality traits modeled after the Myers-Briggs Type Indicator (Myers et al., 1998), with its emotional reactions and facial expressions governed by a memory robot system in a previous work. This system employs prompting with GPT-4o-mini to instantiate personality and emotion frameworks, thereby enabling contextually aware dialogue generation that ensures consistency of personality across diverse conversational topics. In the current study, the high-affective (ENFP) personality was operationalized through prompt

guidelines that prioritized emotional resonance over factual reasoning, expressed care for the user's emotional state during decision-making interactions, and adopted an intuitive, narrative-driven response style. Conversely, the low-affective (ENTP) personality was instantiated through guidelines emphasizing logical analysis and factual accuracy, maintaining evaluative distance from emotional content, and employing an analytical, efficiency-oriented response style.

This explicit prompt-level operationalization systematically incorporated other-directed affective verbal behavior categories, specifically outward emotional resonance and concern for others' emotions (Myers et al., 1998; Plank & Hovy, 2015), within the ENFP condition. Based on the design of the prompts, the linguistic features of the T-F subscale can be further delineated into three distinct dimensions: (1) non-verbal behaviors, including turn-taking, word count, and duration; (2) affective features, encompassing valence, rate, and content; and (3) linguistic style, characterized along a narrative vs efficiency dimensions.

In designing the robot's utterances, a high-affective robot's inquiry was constructed as follows: "I thought we had a very pleasant conversation; sharing our mutual interests brings me joy. I am interested in your passion for dancing. How did you come to develop your love for it?" Conversely, a low-affective robot's typical statement was: "To be honest, financial issues merit particular attention in an aging society. For example, funding strategies for long-term care should be a priority." These utterances were originally in traditional Chinese. The corresponding English translations, along with metric scores for valence, affect, lexical density, cognitive processes, and personal pronoun usage, are provided in Table 2.

Table 2.

Typical utterance examples in high- and low-affective robot design.

Design	Language	Utterance	1	2	3	4	5
High Affective	English (Translated)	Wow, thanks for saying that! I am doing great, ha-ha! I think everyone has their own advantage. Do you have anything you want to do recently?	1.38	7	0.43	0.12	0.12
Low Affective	English (Translated)	To be honest, financial issues are worth special consideration in an aging society. For instance, funding plans for long-term care must be prioritized.	-0.72	0	0.65	0.35	0.04

Note: 1 = valence, 2 = affect, 3 = lexical density, 4 = cognitive process, 5 = personal pronouns usage.

To validate that the two robotic conditions produced linguistically distinguishable stimuli, five linguistic features were extracted from LIWC scores (Lin et al., 2014; Pennebaker et al., 2015) derived from the robots' conversational turns. Affective valence was operationalized as the difference between positive and negative emotional valence, calculated as the z-score of 'pos' minus 'neg'. Affective rate was represented by the ratio of affective words to the total word count, while affective content was quantified by the absolute frequency of affective category words. To differentiate narrative from efficiency-oriented linguistic styles, lexical density was computed as the proportion of lexical words (nouns, verbs, adjectives, and adverbs) (Gregori-Signes & Clavel-Arroitia, 2015), calculated by subtracting function-word counts from the total word count and dividing by the total word count. Additionally, the use of personal pronouns (ppron rate, defined as the ratio of personal-pronoun words to the total word count) and cognitive process categories (cogproc rate, defined as the ratio of cognitive-process words to

the total word count) was analyzed to verify the degree of alignment between prompt design and the resulting linguistic features.

4. Results

4.1. Linguistic Validation of robot stimuli

Utilizing the linguistic feature measures outlined in Section 3, a comparison was conducted between the two robot conditions to assess stimulus distinguishability. As indicated in Table 3, no significant differences were observed in interaction duration or turn count. Nevertheless, consistent with the experimental prompt design, the high-affective robot demonstrated a more narrative verbal style, characterized by lower lexical density, a higher number of utterances, and a higher rate of personal pronouns. Additionally, it produced a greater absolute amount of affective content, despite no significant differences in affective rate or valence compared with the low-affective condition (see Table 4).

As shown in Table 4, to enhance the rigor of our linguistic analyses and account for potential carry-over effects inherent in within-subject experimental designs, we explicitly incorporated the presentation order and its interaction with the robot condition into our Linear Mixed Models (LMMs). Although most linguistic markers did not demonstrate sequence dependency, a significant interaction effect was identified for cognitive processing words ($p_{\text{int}} = 0.003$). This finding suggests a priming effect in conversational settings: the extent to which the low-affective robot employed logical-reasoning language appeared to be partially influenced by prior exposure to the high-affective robot’s narrative style.

Table 3.

Robot condition comparisons on non-verbal features as stimulus validation.

Non-verbal feature	High-Affective		Low-Affective		W	p	d
	M	SD	M	SD			
Turns	45.03	4.83	46.60	3.96	97.50	0.133	0.26
Word Count	1293.30	158.10	1207.30	140.70	144.50	0.07	-0.38
Utterance	28.79	2.73	26.01	3.10	92.00	0.003**	-0.68
Duration (minutes)	15.36	2.06	15.72	3.20	232.00	1.00	0.10

Note: $N = 30$ paired sessions (60 interactions). W = Wilcoxon signed-rank statistic; d = Cohen’s d . Shapiro–Wilk test p-value for duration: 0.0083.

Table 4.

Robot condition comparisons on linguistic features as stimulus validation.

Linguistic feature	High-Affective		Low-Affective		β	p_{holm}	p_{int}	d
	M	SD	M	SD				
Affective Content	2.84	2.06	2.54	1.84	-.342	.002	.683	-.177
Affective Valence	0.05	1.51	-0.50	1.61	-.131	.214	.639	-.085
Affective Rate	0.10	0.07	0.10	0.07	-.001	.829	.987	-.011
Lexical Density	0.58	0.11	0.60	0.14	.025	<.001	.304	.205
ppron rate	0.11	0.05	0.09	0.05	-.019	<.0001	.770	-.376
cogproc rate	0.20	0.09	0.21	0.09	.022	<.0001	.003	.243

Note: $N = 2,749$ turns across 60 sessions. Models were fitted using LMMs with a random intercept for participants to account for within-subject variance. β = fixed-effect estimate; p_{holm} = Holm-Bonferroni-corrected p-value for main effects; p_{int} = uncorrected p-value for the interaction between robot condition and presentation order; d = Cohen's effect size. The high-affective robot serves as the reference group; thus, a positive b indicates a higher score in the low-affective robot condition.

Importantly, even when controlling for this interaction, the primary effect of the robot's cognitive processing of words remained statistically significant ($p_{\text{holm}} < 0.0001$). These results corroborate the finding that the low-affective robot's propensity for analytic and efficiency-oriented communication is a stable characteristic, impervious to variations in presentation order.

As illustrated in Figure 1, the high-affective and low-affective robots exhibit distinct, complementary patterns across eight linguistic octants. The high-affective robot's dialogue predominantly resides within the on-design region on the left, particularly within the "High Affect, Interpersonal, Narrative" category, which accounts for 25.0% of its turns and represents its largest octant. In contrast, the low-affective robot's engagement in this same octant is only 13.2%, less than half that of the high-affective robot. Conversely, the low-affective robot primarily occupies the right on-design region, labeled "High Affect, Impersonal, Efficient," with 24.2%, forming its main octant, while the high-affective robot's presence in this cell is 17.2%. This pattern indicates that the high-affective robot tends to engage more in interpersonal, narrative communication, whereas the low-affective robot favors impersonal, efficient interactions, highlighting their complementary affective and communicative styles.

The high-affective robot allocates 34.5% of its dialogue to the "Interpersonal, Narrative" cluster, while the low-affective robot dedicates 37.0% to "Impersonal, Efficient." This demonstrates a symmetrical distribution within their respective on-design regions. Single-cell analysis reveals that the "Low Affect, Impersonal, Efficient" octant comprises 12.8% of the low-affective robot's turns versus 6.0% for the high-affective robot, a 2.1-fold difference. Conversely, the "Low Affect, Interpersonal, Narrative" octant accounts for 9.5% of high-affective robot turns, surpassing the 6.7% of the low-affective robot. Furthermore, in the context of "off-design" octants, cells characterized by low affect that are aligned with a single design dimension, such as "Low Affect, Impersonal, Narrative," remain marginal, accounting for 3.3% in the high-affective robot condition and 4.7% in the low-affective robot condition. These cells do not predominate in either distribution.

Collectively, the manipulation check validated the linguistic expressions of the two robots. The results corroborate that, despite exhibiting overlap across certain marginal dimensions, the dialogues of each robot predominantly cluster within distinct design octants. This delineation validated the existence of two linguistically distinguishable conversational stimuli.

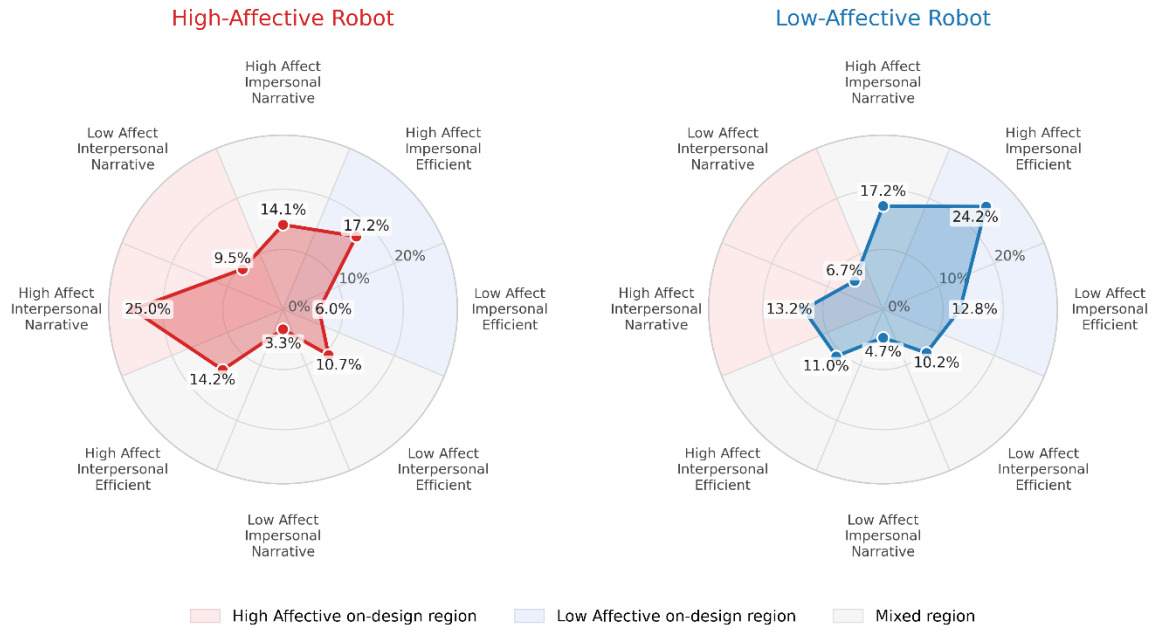


Figure 1. Compositional asymmetry of high-affective robot and low-affective robot dialogue across eight linguistic octants (affective content, personal pronouns and lexical density).

Note: Sectors shaded in pink mark the high-affective on-design region (interpersonal and narrative); blue marks the low-affective on-design region (impersonal, efficient); grey marks mixed octants. Each polygon reports the percentage of conversational turns that fall into each octant for the corresponding robot.

4.2. Identifying affective verbal cue types: Factor structure and scale validation

Table 5 presents the factor structure, including characteristic descriptions and representative items for each subscale. The final instrument consists of 14 items partitioned into three subscales. Based on the identified factor structure, the first component is designated as Feeling Type I (7 items; Cronbach's $\alpha = 0.825$), characterized by outward emotional resonance and concern for others' feelings; the second component is labeled Feeling Type II (4 items; $\alpha = 0.767$), characterized by self-doubt and emotional volatility; and the third component is termed Thinking (3 items; $\alpha = 0.705$), distinguished by a focus on factual information and decisional efficiency. These labels are consistently employed in subsequent analyses. The remaining five items did not meet the criteria for retention and were therefore excluded.

Principal Component Analysis (PCA) of MBTI affectiveness scale responses ($N = 60$ observations from 30 participants $\times$ 2 robot conditions) identified three distinct factors. The preliminary diagnostic tests demonstrated that the data were appropriate for factor extraction (KMO = 0.69, Bartlett's test = 450.82, $p < 0.001$). The three-factor solution explained 49.03% of the variance, which is typical for exploratory factor analyses of newly developed instruments, and aligns with the exploratory nature of the Feeling Type I/II distinction emphasized throughout the study.

Validation analyses confirmed the differential validity of the two exploratory feeling-type subscales in the context of single-session HRI. Utilizing Feeling Type I, which pertains to other-directed emotional resonance, as the classification criterion, 24 out of 30 participants (80%) accurately identified the robot exhibiting high affectivity. Furthermore, the classification of each session according to its Feeling Type I score was compared with participants' qualitative descriptions. The analysis revealed a concordance in 19 out of 29 cases for each robot (66%; one participant did not provide an open-ended description). Fisher's exact test

demonstrated a statistically significant association between classification and description for the low-affective robot ($p = 0.043$), whereas no significant association was observed for the high-affective robot ($p = 0.198$). These findings suggest that the Feeling Type I subscale is partially aligned with participants' holistic impressions of the robots.

Feeling Type II, characterized by self-referential emotional volatility, exhibited limited discriminatory capacity between experimental conditions. As shown in Table 6 and Figure 2(A), at the subscale level, mean scores did not significantly differ between robot conditions (high-affective: $M = 3.81$, $SD = 1.29$; low-affective: $M = 3.82$, $SD = 1.41$; $W = 134.5$, $p = 0.915$). Furthermore, Feeling Type II classifications were unrelated to participants' subjective descriptions: concordance was low for both robots (high-affective: 14 of 29; low-affective: 13 of 29), and Fisher's exact tests found no association between the two classification sources ($p = 1.000$ for both robots). Figure 2(B) plots each participant's Feeling Type I score under the two conditions. The substantial overlap of individual profiles indicates that the absence of a condition effect reflects genuine equivalence at the subscale level rather than the averaging-out of opposing subgroups. At the item level, no individual item differed significantly between conditions; descriptively, Q15 ("This robot is controlled more by emotions than controlling emotions") was rated somewhat higher for the low-affective robot ($M = 4.10$) than for the high-affective robot ($M = 3.80$, $p = 0.299$). Such findings support the interpretation that Feeling Type II primarily encapsulates emotional instability rather than warmth, and that it fails to effectively convey affective qualities as originally intended.

Table 5.

MBTI-based affectiveness scale: factor structure and item composition.

Factor	Characteristic	Representative items
Feeling Type I (7 items)	Outward emotional resonance; care for others' feelings; intuitive decision-making	Q16: Cares more about others' feelings than logic; Q3: Finds personal stories more persuasive than data; Q19: Relies on emotions and intuition rather than logical reasoning (Q13 removed after item analysis)
Feeling Type II (4 items)	Self-doubt; emotional volatility; worry; rumination over past mistakes	Q4: Small mistakes cause self-doubt; Q9: Mood may change rapidly; Q12: Often worries things will turn worse; Q14: Troubled by past mistakes (Q2, Q10, Q11 removed after item analysis)
Thinking (3 items)	Prioritization of facts; logical efficiency; analytic style	Q5: Prioritizes facts over feelings; Q7: Values decision-making efficiency; Q8: Proves own point before considering others' feelings (Q6 removed after item analysis)
Removed after item analysis (5 items)	Items with negative or near-zero CITC indicating content misalignment with subscale theme	Q2, Q6, Q10, Q11, Q13

Note: CITC = Corrected Item-Total Correlation. Items with CITC < 0.30 or negative CITC were removed. α = Cronbach's alpha.

Table 6.

Wilcoxon signed-rank tests: robot condition comparisons on sub-scale scores.

Subscale	High-Affective		Low-Affective		W	p	r
	M	SD	M	SD			

Feeling Type I (other-directed)	4.33	1.25	4.06	1.16	132	.171	.250
Feeling Type II (self-referential)	3.81	1.29	3.82	1.41	134.5	.915	.019
Thinking	3.09	1.22	3.34	1.47	129	.149	.263

Note: $N = 30$. Comparisons are paired (within-subjects). W = Wilcoxon signed-rank statistic; r = effect size ($|z|/\sqrt{N}$). Scale range: 1–7.

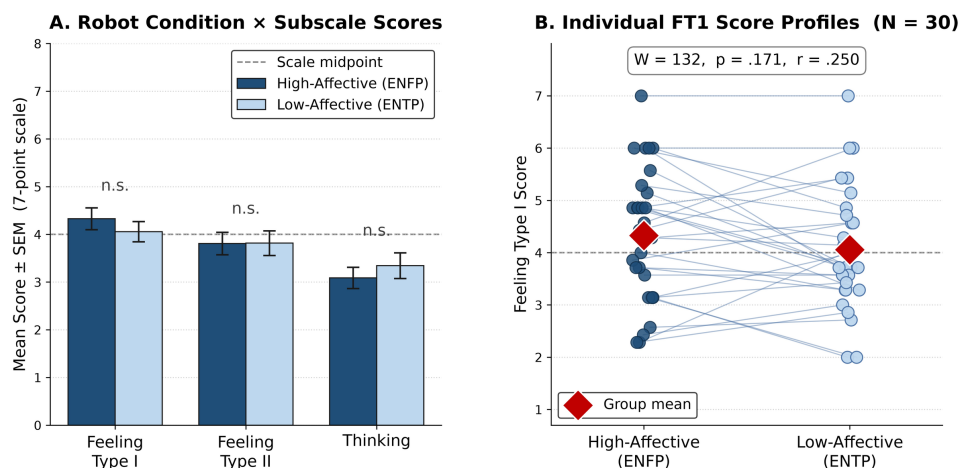


Figure 2. Perceived robot subscale scores by condition.

Note: Panel A: mean ($\pm$ SEM) scores on Feeling Type I, Feeling Type II, and Thinking subscales (range 1–7) for the high-affective (ENFP) and low-affective (ENTP) robots; n.s. = not significant. Panel B: individual participants' Feeling Type I score profiles across conditions ($N = 30$); red diamonds indicate condition means. No subscale differed significantly between conditions (Wilcoxon signed-rank tests, all $p > .05$; see Table 6).

Analysis indicates that outward emotional resonance (Feeling Type I) most effectively discriminates this sample, suggesting it encompasses care for others' feelings and intuitive responses. Conversely, Feeling Type II behaviors of self-doubt, volatility, and rumination conflict with older adults' expectations of robot stability, and should be excluded from affective robot design for this population.

4.3. Descriptive statistics and interaction engagement

Consistent with the duration equivalence shown in Table 3, participants engaged in prolonged interactions with both robots, averaging 15.54 min per session ($SD = 2.67$; range: 11.90–26.20 min), as recorded via system logs. Interaction with the high-affective robot lasted an average of 15.36 min ($SD = 2.06$; range: 12.27–20.92), while the low-affective robot interaction averaged 15.72 min ($SD = 3.20$; range: 11.90–26.20). A Linear Mixed-Effects Model (LMM), with robot personality and interaction order as fixed effects and participant ID as a random effect, indicated that robot personality did not significantly influence interaction duration ($\beta = 0.45$, $SE = 0.62$, $p = 0.46$), after controlling for session order ($\beta = -1.29$, $p = 0.038$). These findings suggest the robustness of interaction length across different sequences and individual differences. Regarding preferences, 43.3% ($n = 13$) favored the high-affective robot, 36.7% ($n = 11$) favored the low-affective robot, and 20.0% ($n = 6$) had no preference (see Table 9). Descriptive statistics for primary measures are provided in Table 7.

To ensure that the counterbalanced presentation order did not introduce systematic bias, robot presentation order (high-affective-first, $n = 16$; low-affective-first, $n = 14$) was included as a between-subjects factor in a mixed ANOVA alongside the within-subjects robot condition.

Results revealed no significant main effects of presentation order on Feeling Type I affectiveness scores ($F(1, 28) = 0.340, p = 0.565, \eta^2_p = 0.012$), nor of robot condition ($F(1, 28) = 2.210, p = 0.148, \eta^2_p = 0.073$), and no significant interaction ($F(1, 28) = 0.007, p = 0.934, \eta^2_p < 0.001$). Robot preference distributions also did not differ significantly across presentation order ($\chi^2 = 2.896, p = 0.235$). Cell means for Feeling Type I ranged from 3.94 to 4.45 across conditions and orders, indicating comparable baselines. These findings suggest that differences between robot conditions reflect genuine perceptual distinctions, supporting the internal validity of the within-subjects design. The mixed ANOVA results are provided in Table 8.

Table 7.
Descriptive statistics for primary scale variables (N = 30).

Variable	M	SD	Mdn	Min	Max
Feeling Type I — High-Affective Robot	4.33	1.25	4.36	2.29	7.00
Feeling Type I — Low-Affective Robot	4.06	1.16	3.71	2.00	7.00
Feeling Type II — High-Affective Robot	3.81	1.29	3.88	1.75	7.00
Feeling Type II — Low-Affective Robot	3.82	1.41	3.62	1.25	6.75
Thinking — High-Affective Robot	3.09	1.22	3.00	1.00	6.00
Thinking — Low-Affective Robot	3.34	1.47	3.33	1.00	6.33
BFI-15 Agreeableness	12.33	1.86	12.00	7.00	15.00
BFI-15 Openness	10.97	2.55	11.00	6.00	15.00
BFI-15 Conscientiousness	12.83	1.88	13.00	7.00	15.00
BFI-15 Extraversion	12.23	2.22	12.00	5.00	15.00
BFI-15 Neuroticism	8.47	3.05	8.00	3.00	15.00
TAM Total Score	64.10	11.41	64.50	23.00	77.00

Note: FT = Feeling Type; BFI-15 = Big Five Inventory-15; TAM = Technology Acceptance Model. Scale scores: Feeling Type I and II, Thinking (1–7); BFI-15 subscales (3–15); TAM total (11–77). *Mdn* = Median.

Table 8.
Mixed ANOVA results: Presentation order effects on feeling type I scores.

Source	SS	df1	df2	MS	F	p	η^2_p
Presentation Order	0.841	1	28	0.841	0.340	.565	.012
Robot Condition	1.105	1	28	1.105	2.210	.148	.073
Order $\times$ Condition	0.004	1	28	0.004	0.007	.934	<.001
Error (Between)	23.495	28		0.839			
Error (Within)	13.668	28		0.488			

Note: Between-subjects factor: Presentation Order (high-affective-first, $n = 16$; low-affective-first, $n = 14$). Within-subjects factor: Robot Condition (high-affective vs. low-affective). η^2_p = partial eta-squared. No significant effects were observed.

Table 9.
Robot preference distribution by agreeableness group.

Agreeableness group	High-Affective	No preference	Low-Affective	Total
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	(ENFP)		(ENTP)	
High (BFI-A $\geq$ 12, $n = 22$)	12 (54.5%)	2 (9.1%)	8 (36.4%)	22
Low (BFI-A $<$ 12, $n = 8$)	1 (12.5%)	4 (50.0%)	3 (37.5%)	8
Total	13 (43.3%)	6 (20.0%)	11 (36.7%)	30

Note: $N = 30$. Agreeableness groups defined by BFI-15 Agreeableness median ($Mdn = 12$). Percentages within rows. $\chi^2(2) = 7.305$, $p = 0.026$, Cramer's $V = 0.493$.

4.4. Linguistic features and the three explorative perception factors

As the high- and low-affective robots produce linguistically distinct utterances across affective content, lexical density, and pronoun use, further examinations were conducted to confirm whether these linguistic differences correspond, at the session level, to the three exploratory factors derived from participant ratings: Feeling Type I (FT-I), Feeling Type II (FT-II) and Thinking. The analysis evaluates this correspondence from both perspectives: robot output and user response, assessing the degree of linguistic-perceptual coupling resulting from the manipulation.

4.4.1. Session-level correspondence between LIWC features and perception factors

For each of the 60 sessions (comprising 30 participants under two conditions), all robot turns were aggregated to generate a session-level LIWC profile encompassing 80 linguistic features, including the LIWC2015 Traditional Chinese categories and the derived Lexical Density metric. Subsequently, correlations were computed between each feature and the participants' three subscale scores. Analyzing the resulting 240 LIWC-feature and factor pairs, none exhibited a statistically significant correlation after applying the Holm correction at $\alpha = .05$. The most notable descriptive trends were uniformly modest: the auxiliary-verb rate demonstrated a negative correlation with FT-I ($r = -0.366$, $p_{\text{holm}} = 0.96$), the positive-emotion rate showed a negative correlation with Thinking ($r = -0.313$, $p_{\text{holm}} = 1.00$), and the cognitive-process rate was negatively correlated with FT-I ($r = -0.268$, $p_{\text{holm}} = 1.00$). A LassoCV feature selection process per factor (employing 5-fold cross-validation and standardized inputs) identified a single feature for FT-I (auxverb rate, $\beta = -0.057$), while no features were selected for FT-II or Thinking. Given the sample sizes ($n = 60$ versus approximately 80 features), these results suggest that no individual LIWC feature of the robot's output serves as a robust per-session predictor of any of the three perception factors. The complete 240-pair correlation matrix is available in the study's data deposit.

4.4.2. Symmetry check on the user side

To evaluate whether the same correspondence persists when the linguistic signal is sourced from participants' own utterances, an identical session-level pipeline was implemented on user turns (CKIP-segmented, LIWC-coded, and aggregated per session; $n = 60$). Two analytical approaches were employed: (a) paired t-tests comparing F versus T on each LIWC metric, and (b) Pearson correlation analyses between LIWC rates and three perception factors. Neither approach revealed effects that remained significant after Holm correction. The most prominent trend observed was an increased rate of personal pronouns in the high-affective condition (ppron: $M = 0.108$ vs. 0.095 ; $t = 2.66$; $p_{\text{raw}} = 0.013$; $p_{\text{holm}} = 1.00$), indicating a modest mirroring of the robot's interpersonal style. Regarding correlations, the most notable association was a negative correlation between adverb usage and the Thinking perception factor ($r = -0.366$, $p_{\text{holm}} = 0.97$). A Lasso cross-validation performed on user-side features resulted in the selection of no features for FT-I or FT-II, and a single feature for the Thinking factor, specifically, adverb usage ($\beta = -0.031$).

4.4.3. Integration: Turn-level asymmetry, session-level convergence to null

Both session-level analyses yielded consistent results: neither the session-aggregated LIWC profiles of the robot nor those of the participant predicted the three perception factors after correction for multiple comparisons. Additionally, the condition contrast on the user-side language remained nonsignificant. This finding contrasts with the turn-level linguistic asymmetry documented in Figure 1 and Table 2, where the two conditions showed reliable differences on marginal LIWC dimensions such as affective content, lexical density, and pronoun usage. The overall pattern suggests that when linguistic signals are aggregated across multiple turns and aligned with the participant’s overall subscale impression, the per-cue signal falls below the detection threshold in an $n = 30$ within-subject design. This holds true regardless of whether LIWC features are derived from the robot’s output or the participant’s responses. Consequently, the robust turn-level linguistic asymmetry does not translate into a measurable, per-factor session-level effect within this sample.

4.5. Agreeableness and robot affectiveness preferences

Multinomial logistic regression examined how Big Five personality traits influenced robot preferences across three categories: high-affective, low-affective, or no preference. Among all personality dimensions, agreeableness emerged as the sole significant predictor of preference patterns. Participants preferring the high-affective robot scored highest in agreeableness ($M = 13.23$), followed by those preferring the low-affective robot ($M = 11.82$), and those expressing no preference ($M = 11.33$).

Agreeableness was a significant predictor of both preference for low-affective robots ($\beta = -2.62, p = 0.048$) and the absence of preference ($\beta = -3.95, p = 0.020$). In contrast, all other Big Five personality dimensions demonstrated non-significant effects. The negative coefficients suggest that greater agreeableness is associated with a decreased likelihood of favoring non-emotional robot interactions: a one-point increase on the BFI-15 agreeableness scale corresponded to an approximate 76% reduction in the odds of preferring the low-affective robot, and an approximately 88% reduction in the odds of expressing no preference. Standardized coefficients and confidence intervals are detailed in Table 10. A simplified agreeableness-only model demonstrated superior fit relative to the full five-predictor model (AIC: 64.45 vs. 67.93; BIC: 70.06 vs. 84.74), confirming that other personality traits contributed negligible predictive value. Bootstrap analysis across 2,000 iterations confirmed robust effects despite the modest sample size, with effect sizes indicating large practical significance (Cohen's $d = 0.856$ for high vs. low affective preferences; $d = 1.158$ for high-affective vs. no preference).

Table 10.
Multinomial logistic regression: Big Five predictors of robot preference.

Predictor	β	SE	Wald	p	OR	95% CI
<i>No Preference vs. High-Affective (reference)</i>						
Openness	0.565	0.856	0.436	.509	1.760	[0.33, 9.41]
Conscientiousness	1.425	1.097	1.688	.194	4.157	[0.48, 35.66]
Extraversion	1.796	1.226	2.147	.143	6.027	[0.55, 66.62]
Agreeableness	-3.951	1.695	5.432	.020*	0.019	[0.00, 0.53]
Neuroticism	0.992	0.714	1.933	.164	2.697	[0.67, 10.92]
<i>Low-Affective vs. High-Affective (reference)</i>						

Openness	1.230	0.723	2.896	.089	3.422	[0.83, 14.11]
Conscientiousness	0.970	0.951	1.040	.308	2.637	[0.41, 17.01]
Extraversion	0.132	0.909	0.021	.884	1.141	[0.19, 6.78]
Agreeableness	-2.616	1.325	3.899	.048*	0.073	[0.01, 0.98]
Neuroticism	-0.082	0.544	0.023	.880	0.921	[0.32, 2.67]

Note: $N = 30$. Reference category = High-affective preference. All predictors standardized ($M = 0$, $SD = 1$). AIC = 67.93; BIC = 84.74. * $p < 0.05$.

Regarding behavioral tendencies, among participants characterized by high agreeableness ($BFI-A \geq 12$; $n = 22$), 54.5% ($n = 12$) preferred the high-affective robot. In contrast, only 12.5% ($n = 1$) of participants with low agreeableness ($BFI-A < 12$; $n = 8$) demonstrated such a preference, indicating that personality-based preference patterns are consistently evident in human-robot interaction contexts when fundamental interaction protocols are adhered to. Table 9 presents the full distribution of preferences across agreeableness groups, while Table 10 reports the comprehensive multinomial logistic regression results.

5. Discussion and conclusion

5.1. Feeling type I as the primary design principle for affective single-session HRI

The central empirical contribution of this study is the identification of two qualitatively distinct affective verbal cue subtypes that have opposite effects on affectiveness perception in single-session HRI with older adults. Understanding why Feeling Type I succeeds while Feeling Type II fails requires engaging with the intersection of thin-slice judgment theory and older adults' HRI interaction scripts.

Linguistically, the cue categories of Feeling Type I and Feeling Type II elicited distinguishable dialogue patterns; however, their causal influence on participants' overall subscale ratings was not statistically significant at the session level. This suggests that multi-turn perception involves an integration of individual cue categories, a process that remains unresolved within the constraints of a small sample size ($n = 30$) in a within-subject design.

Feeling Type I behaviors are fundamentally outward-directed. They encompass prioritizing others' emotional states in decision-making, expressing empathic resonance, and using intuitive rather than analytic reasoning. They involve attending to and responding to the user's emotional experience, a behavioral orientation that is directly legible through conversational cues within a short interaction. From the perspective of thin-slice behavioral analysis, these behaviors constitute high-visibility social cues analogous to the warmth dimension that Carpinella et al. (2017) identified as universally accessible in brief robot evaluations. They satisfy older adults' core interaction script expectation of emotional responsiveness without requiring extended observation, explaining why Feeling Type I was discriminated successfully within a single 15-minute session.

Feeling Type II behaviors are fundamentally inward-directed. They encompass expressions of self-doubt, mood volatility, worry about outcomes, and rumination over past mistakes. Rather than signaling attention to the user, they signal the robot's own emotional vulnerabilities. What appears as emotional openness in human-human interaction carries an altogether different meaning in the HRI context: these same behaviors violate a foundational expectation in older adults' robot interaction scripts: namely, that robots should be stable, reliable, and competent. In cultural contexts where stability and precision are expected from technological agents (and where Confucian values further emphasize emotional self-regulation and propriety), Feeling

Type II cues are likely to trigger evaluations of the robot as malfunctioning or psychologically unstable rather than emotionally warm. The descriptive tendency for Q15 (‘controlled by emotions rather than controlling emotions’) to be rated more highly for the low-affective robot than the high-affective robot, although not statistically significant, is consistent with this interpretation: Feeling Type II cues were perceived as emotional dysregulation, an attribute that older adults associated with the condition they rated as less affective overall.

These findings hold direct implications for prompt engineering of LLM in companion robot design. Robot designers should prioritize Feeling Type I verbal behaviors, including expressing empathic interest in the user’s stories and feelings, using emotionally resonant language, and framing responses through the lens of the user’s experience rather than abstract principles. Feeling Type II behaviors should be explicitly excluded from affective robot prompts: expressions of self-doubt, statements about mood changes, and references to past mistakes or worry are likely to undermine rather than enhance perceived affectiveness regardless of the robot’s overall personality design. This provides a practical, actionable distinction that can be operationalized directly as a prompt design guideline for LLM-driven social robots. Recent co-design research within the IHCI community has highlighted the importance of grounding companion robot design in empirically characterized user expectations across natural interaction contexts (Ahmed et al., 2026).

The distinction between Feeling Types I and II offers two significant contributions to theoretical understanding that merit explicit articulation. First, it operationalizes within a human-robot interaction context, an application novel to our knowledge, the conditions under which affective verbal cues become amenable to thin-slice extraction: specifically, cues must be directed toward others and perceptible in a relational context, as opposed to being self-referential or emotionally volatile. While the thin-slice judgment literature has established that observers can form rapid and accurate impressions from limited behavioral samples, it has not specified which properties of these cues facilitate such extraction during human-robot interactions. The present findings elucidate this issue in the realm of verbal behavior design. Second, the distinction reveals that the MBTI Feeling dimension, typically regarded as a unidimensional construct in robot personality research and design, comprises functionally asymmetric subtypes whose divergent perceptual effects cannot be inferred solely from their shared dimensional label. In this context, emotional richness and emotional appropriateness emerge as orthogonal design considerations: a robot script may be high in affective content but still counterproductive if that content is directed inwardly. Collectively, these contributions shift the design paradigm for LLM-based companion robots from an emphasis on maximizing emotional expression to a nuanced calibration of the directionality of that expression.

5.2. Agreeableness as an individual-difference moderator of affective cue sensitivity

The finding that agreeableness was the sole significant Big Five predictor of robot affectiveness preference extends the established relationship between agreeableness and interpersonal empathy to the human-robot interaction context, providing support for applying the CASA paradigm to older adult populations. Despite age-related changes in cognitive processing and technology experience, the fundamental personality-driven social preference pattern transferred consistently from human-human to human-robot interaction when basic interaction script requirements were met. Specifically, highly agreeable individuals showed enhanced sensitivity to others’ emotional states in the HRI context, replicating a pattern well-established in human-human interaction research. This aligns with recent evidence that social disposition moderates trust formation and response differentiation across robot interaction conditions (Yueh et al., 2026b; 2026c), suggesting that agreeableness functions as a broader sensitivity moderator in HRI beyond the specific context of affectiveness perception.

The selectivity of agreeableness among the Big Five dimensions is theoretically informative. The finding that extraversion, conscientiousness, openness, and neuroticism showed no significant effects suggests that robot affectiveness specifically activates the empathy-related personality systems associated with agreeableness, rather than broader social engagement or openness to experience. From a thin-slice perspective, this makes theoretical sense: Feeling Type I behaviors are precisely the type of outward, other-directed emotional cues that the empathy-enhanced sensitivity of highly agreeable individuals should detect most readily. The connection between agreeableness and Feeling Type I sensitivity is thus not merely correlational but theoretically grounded. High-agreeableness individuals are better positioned to rapidly detect and positively evaluate the outward emotional resonance that constitutes Feeling Type I, and are less likely to be deterred by its absence in low-affective robot interactions.

The agreeableness finding has a practical corollary: systems capable of rapid personality assessment could use agreeableness scores to calibrate the intensity of Feeling Type I verbal behaviors in real time. Such assessment is achievable through brief conversational interactions or validated short-form questionnaire instruments. This calibration would yield more emotionally resonant responses for high-agreeableness users while maintaining a neutral, competence-focused communication style for low-agreeableness users who may find excessive emotional expressiveness uncomfortable or unnecessary.

5.3. The 20% nonidentifiers: Implications for adaptive robot design

While 80% of participants successfully identified the high-affective robot's personality, the remaining 20% were unable to do so. This result points toward design consequences that extend well beyond its role as a methodological limitation. Current robot design approaches typically assume successful personality perception, calibrating affectiveness levels to a target personality type and expecting users to respond accordingly. The present data suggest that a substantial minority of older adult users may not reliably differentiate robot personalities within a single conversational session, and that systems designed to adapt based on personality perception may therefore fail to engage this group as intended.

Future research should examine the individual-level factors that predict successful versus unsuccessful discrimination. Candidate moderators include cognitive flexibility, prior technology exposure, hearing acuity, and personality dimensions not captured in the present study (such as openness to experience, which may influence willingness to anthropomorphize novel agents). From a design perspective, these findings argue for developing fallback interaction strategies that do not depend on personality-based affectiveness cues. For instance, adaptive systems could monitor behavioral engagement markers as real-time indicators of interaction quality, including response latency, elaborateness of replies, and topic initiation frequency, independent of whether the user explicitly perceives the robot's personality type.

5.4. Cultural context and generalizability

This study, conducted with Taiwanese older adults in a single-session Mandarin context, highlights cultural influences on emotional expression, particularly the failure of Feeling Type II. Confucian values emphasizing emotional self-regulation, social harmony, and face maintenance heighten sensitivity to emotional dysregulation indicators. A robot displaying self-doubt, mood swings, or rumination contravenes culturally internalized norms of appropriate emotional conduct among older Taiwanese adults. The pronounced failure of Feeling Type II likely reflects the high-uncertainty avoidance and Confucian heritage, contrasting with Western individualist contexts where emotional vulnerability signifies authenticity rather than instability.

The effectiveness of Feeling Type I may be broadly generalizable, but this requires qualification. Its core behaviors of expressing care for the user's feelings, emphasizing emotional resonance, and considering the user's experiential perspective are manifestations of the warmth dimension. Warmth is universally central to social evaluation across cultures (Fiske et al., 2007). If warmth traits are cross-culturally consistent, their observable verbal behaviors may also generalize. However, the specific forms these behaviors take, such as conversational tone, topics appropriate for emotional expression, and levels of explicitness, are culturally shaped. Therefore, cross-cultural replication is essential to determine if Feeling Type I's verbal patterns extend beyond the Taiwanese Mandarin context.

5.5. Limitations and future directions

Several limitations warrant consideration when interpreting the present findings. The sample size of $N = 30$, while statistically adequate for the within-subjects design and bootstrap-validated for the regression analyses, is modest for multinomial logistic regression and limits the statistical precision of the preference pattern estimates. The marginal KMO value of 0.69 introduces uncertainty regarding the stability of the three-factor solution across samples, and the Feeling Type I/II distinction should be treated as an empirically derived hypothesis requiring replication with larger samples and alternative measurement approaches before being adopted as an established construct. The single-session design, while appropriate for the thin-slice judgment research questions addressed here, cannot speak to whether the identified preferences are stable over extended interaction periods or whether they would be modified by repeated exposure and relationship formation over time.

Anthropomorphism in human-robot interaction serves as a double-edged sword. While it can foster engagement, it also risks creating deceptive emotional attachments, particularly threatening to older adults (Bryson, 2010; Turkle, 2011). Conversely, excessive anthropomorphism may hinder realistic social interactions. Ethical considerations are crucial to evaluate potential relational risks posed by anthropomorphism, which our study addressed successfully without ethical issues. However, gender imbalance limits the generalizability of our findings, especially given that gender influences traits like agreeableness (Costa et al., 2001; Feingold, 1994; Vecchione et al., 2012; Weisberg et al., 2011). Future research should consider gender as a confounding factor. Given the sample's gender imbalance, interpretations of the personality-preference relationships should be cautious.

Finally, multiple-session designs can reveal whether participants develop sensitivity to internal personality features such as emotional rumination over time. Research indicates that novelty effects diminish after 2–8 weeks of interaction (Leite et al., 2013), and social penetration theory suggests that deeper personality layers become accessible only as relationships deepen (Altman & Taylor, 1973). Future research should prioritize replication with larger, more culturally diverse samples to verify if the Feeling Type I/II distinction and the agreeableness-affectiveness relationship extend beyond the Taiwanese context. Longitudinal studies involving repeated interactions could determine if the observed effects remain stable as older adults become familiar with a robot's personality. Additionally, collecting behavioral data such as vocal prosody, response latency, and topic initiation would triangulate self-reports with objective indicators of affective engagement, providing a comprehensive understanding of the mechanisms underlying the effects associated with the Feeling Type I.

5.6. Conclusion

This study offers an exploratory empirical analysis of affective verbal cue types in single-session human-robot interaction with older adults, delineating a theoretically grounded and practically applicable distinction between Feeling Type I (effective) and Feeling Type II

(counterproductive) affective behaviors. The finding that agreeableness uniquely predicts preference for high-affective robots aligns with interpersonal preference patterns documented in human-human interaction research, thereby supporting the extension of the CASA paradigm to older adult populations and highlighting the consistency of personality-driven social preferences across human and artificial agents. These insights provide practical implications for large language model prompt engineers, social robot designers, and gerontechnology researchers engaged in the development of emotionally adaptive companion robots capable of forming appropriate and meaningful affective connections with older adult users within a single interaction session.

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Author contributions

CRedit: Hsiu-Ping Yueh: Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing; Hung-Chun Chang: Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing; Jia-Hsun Lo: Software, Writing – original draft, Writing – review & editing; Shih-Han Wang: Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing; Huan-Yu Chang: Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing; Pei-Shan Yang: Conceptualization, Methodology, Writing – review & editing; Han-Pang Huang: Resources, Supervision, Writing – review & editing.

Ethics statement

This study was approved by the Institutional Review Board of National Taiwan University (IRB approval no.: 202410HS011). All participants provided written informed consent prior to participation and were fully informed of the study purpose, procedures, voluntary nature of participation, and their right to withdraw at any time without penalty or consequence. Data were anonymized prior to analysis.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The de-identified dataset supporting the findings of this research is publicly accessible via the Open Science Framework at <https://doi.org/10.17605/OSF.IO/FB9W8>

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